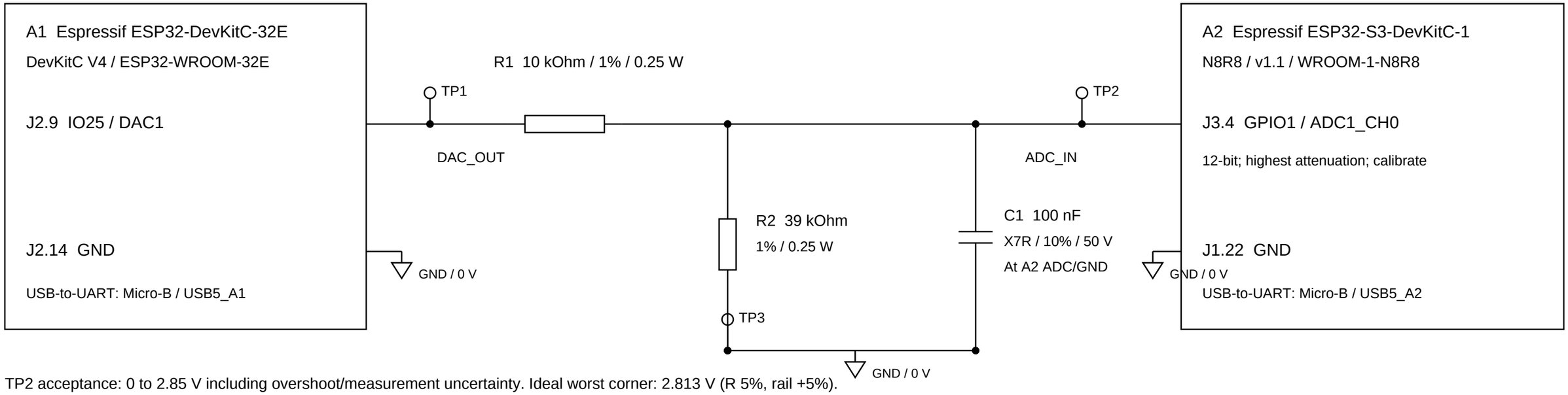
RF-HIL-001 / MODULE INTERCONNECT SCHEMATIC

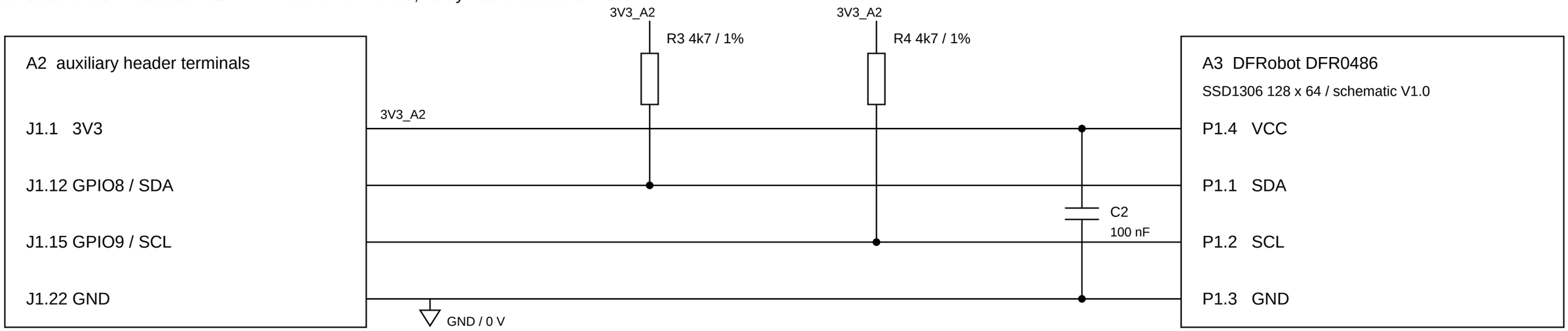
Rev C | 2026-10-21 | Bench prototype | NOT BENCH VERIFIED | Sheet 1 / 1

1. ANALOG PATH - all voltages relative to common signal GND



TP2 acceptance: 0 to 2.85 V including overshoot/measurement uncertainty. Ideal worst corner: 2.813 V (R 5%, rail +5%).

2. OLED / I2C - host-side R3/R4 FITTED for DFR0486; verify rise time and low level



NOTES: dots = junctions; crossings without dots are NOT connected. GND triangles = common signal reference, not protective earth.

A3 numbering follows manufacturer schematic P1, not wiki table row numbers. Confirm silk and cable continuity before power.

C1/C2 non-polarized X7R 100 nF / 10% / 50 V. C2 at A3. R3/R4 are external host pull-ups; A3 contains level shifters/pull-ups.

USB5_A1 and USB5_A2 are separate: do not join 5 V or regulator 3V3 outputs. A3 power only from A2. Check both 3V3 rails +/-5%.

Wire only with both supplies OFF. Power A2 before A1; turn A1 off before A2. No hot-plug protection. No patient/electrode connection.